

44 g/mol

22.4 L/mole

Propane (C_3H_8) is to be burned with 15% excess air. Before entering the furnace, the air is preheated from $0^\circ C$ to $300^\circ C$. If the feed rate of propane is 3.82×10^6 L(STP)/hr, at what rate must heat be transferred (in kJ/hr) to the air in the preheater? Assume complete combustion. Data is on the back of this sheet. At STP ($0^\circ C$ and 1 atm) an ideal gas has a molar volume of 22.4 L/mole.

$$T_1 = 0^\circ C \rightarrow \hat{H}_{air,i} = -0.72$$

$$T_2 = 300^\circ C \rightarrow \hat{H}_{air,f} = 8.17$$

$$\Delta \hat{H} = 8.89$$

$$\dot{n}_{C_3H_8} = 3.82 \times 10^6 \text{ L/hr}$$

$$\text{molar volume } 22.4 \text{ L/mole}$$

$$\text{find kJ/hr}$$

$$\cancel{\Delta U = \Delta E_k + \Delta E_p = Q + W}$$

$$3.82 \times 10^6 \text{ L/hr}$$

$$PV = nRT$$

Atm, L, R

$$\frac{22.4 \text{ L}}{\text{mole}} \times \frac{\text{mol}}{44 \text{ g}} = 0.51 \text{ L/g}$$

$$P = 1$$

$$V = 22.4 \text{ L/mol} \times 44 \text{ g/mol} = 985.6 \text{ L}$$

$$\dot{n} = 3.82 \times 10^6 \text{ L/hr} \times$$

$$R = 0.082057 \frac{\text{atm L}}{\text{gmol K}}$$

$$T = 300^\circ K = 573^\circ K$$

$$n = \frac{PV}{RT} = \frac{(1 \text{ atm})(985.6 \text{ L})}{(0.082057 \frac{\text{atm L}}{\text{gmol K}})(273^\circ K)} = 44 \text{ moles @ } 0^\circ C \text{ STP}$$

$$= 20.96 \text{ moles}$$

← gives C_3H_8 flow rate
need air flow

$$\frac{44 \text{ moles}}{22.4 \text{ L/mole}} = 1.96$$

$$20.96 \text{ for } 300^\circ C$$

$$Q = \Delta \hat{H} = n \Delta \hat{H}_r$$

$$= (44 \text{ moles})(8.89 \text{ kJ/mole})$$

$$\text{rate} = 391.16 \text{ kJ}$$

$$\text{or } (3.82 \times 10^6 \text{ L/hr} / 985.6 \text{ L/mol})$$

$$= 38758.12 \text{ gmol/hr} \times 44 = 1705358$$

$$= 344226 \text{ kJ/hr}$$

$$\text{when } 886.86$$

h... ..

Table B.8 Specific Enthalpies of Selected Gases: SI Units

$\hat{h}$ (kJ/mol)							
Reference state: Gas, $P_{ref} = 1$ atm, $T_{ref} = 25^\circ\text{C}$							
T	Air	O ₂	N ₂	H ₂	CO	CO ₂	H ₂ O
0	-0.72	-0.73	-0.73	-0.72	-0.73	-0.92	-0.84
25	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100	2.19	2.24	2.19	2.16	2.19	2.90	2.54
200	5.15	5.31	5.13	5.06	5.16	7.08	6.01
300	8.17	8.47	8.12	7.96	8.17	11.58	9.57
400	11.24	11.72	11.15	10.89	11.25	16.35	13.23
500	14.37	15.03	14.24	13.83	14.38	21.34	17.01
600	17.55	18.41	17.39	16.81	17.57	26.53	20.91
700	20.80	21.86	20.59	19.81	20.82	31.88	24.92
800	24.10	25.35	23.86	22.85	24.13	37.36	29.05
900	27.46	28.89	27.19	25.93	27.49	42.94	33.32
1000	30.86	32.47	30.56	29.04	30.91	48.60	37.69
1100	34.31	36.07	33.99	32.19	34.37	54.33	42.18
1200	37.81	39.70	37.46	35.39	37.87	60.14	46.78
1300	41.34	43.38	40.97	38.62	41.40	65.98	51.47
1400	44.89	47.07	44.51	41.90	44.95	71.89	56.25
1500	48.45	50.77	48.06	45.22	48.51	77.84	61.09